Experiment No-4
BASIC OPERATIONS ON MATRICES

AIM: Generate a matrix and perform basic operation on matrices using MATLAB software.

Software Required: MATLAB software

Theory:

MATLAB treats all variables as matrices. Vectors are special forms of matrices and contain only one row or one column. Whereas scalars are special forms of matrices and contain only one row and one column. A matrix with one row is called row vector and a matrix with single column is called column vector.

The first one consists of convenient matrix building functions, some of which are givenbelow.
1. eye - identity matrix
2. zeros - matrix of zeros
3. ones - matrix of ones
4. diag - extract diagonal of a matrix or create diagonal matrices
5. triu - upper triangular part of a matrix
6. tril - lower triangular part of a matrix
7. rand - ran

commands in the second sub-category of matrix functions are
1. size- size of a matrix
2. det -determinant of a square matrix
3. inv- inverse of a matrix
4. rank- rank of a matrix
5. rref- reduced row echelon form
6. eig- eigenvalues and eigenvectors
7. poly- characteristic polynomialdomly generated matrix

Program:

```
% Creating a column vector
>> a=[1;2;3]
a =
1
2
3
% Creating a row vector
>> b=[1 2 3]
b=
1 2 3
```

% Creating a matrix

>> m=[1 2 3;4 6 9;2 6 9]

m =

1 2 3

4 6 9

2 6 9

% Extracting sub matrix from matrix

>> sub_m=m(2:3,2:3)

sub_m =

6 9

6 9

% extracting column vector from matrix

>> c=m (:,2)

c =

2
6
6

% extracting row vector from matrix

>> d=m (3,:)

d =

2 6 9

% creation of two matrices a and b

>> a=[2 4 -1;-2 1 9;-1 -1 0]

a =

2 4 -1

-2 1 9

-1 -1 0

>> b=[0 2 3;1 0 2;1 4 6]

b =

0 2 3

1 0 2

1 4 6

% matrix multiplication

>> x1=a*b

x1 =

3 0 8

10 32 50

-1 -2 -5

% element to element multiplication

>> x2=a.*b

x2 =

0 8 -3

-2 0 18

-1 -4 0

% matrix addition

>> x3=a+b

x3 =

2 6 2

-1 1 11

0 3 6

% matrix subtraction

>> x4=a-b

x4 =

2 2 -4

-3 1 7

-2 -5 -6

% matrix division

>> x5=a/b

x5 =

-9.0000 -3.5000 5.5000

12.0000 3.7500 -5.7500

3.0000 0.7500 -1.7500

% element to element division

>> x6=a./b

Warning: Divide by zero.

x6 =

Inf 2.0000 -0.3333

-2.0000 Inf 4.5000

-1.0000 -0.2500 0

% inverse of matrix a

>> x7=inv(a)

x7 =

-0.4286 -0.0476 -1.7619

0.4286 0.0476 0.7619

-0.1429 0.0952 -0.4762

% transpose of matrix a

>> x8=a'

x8 =

2 -2 -1

4 1 -1

-1 9 0

RESULT: Matrix operations are performed using Matlab software.

VIVA QUESTIONS:-

1. Expand MATLAB? And importance of MATLAB?
2. What is clear all and close all will do?
3. What is disp() and input()?
4. What is the syntax to find the eigen values and eigenvectors of the matrix?
5. What is the syntax to find rank of the matrix?